

CHARACTERIZATION OF CRYSTAL EFFICIENCY

*2024 dataset
for TCCP and TCCPA*

JULY 2026
ANNA SOFIA MORO

...

- Aim of reproducing the data analysis described in the article "Performance of short and long bent crystals for the TWOCRIST experiment at the LHC." (*The European Physical Journal*) with the October 2024 dataset for the 2 long crystals TCCP, TCCPA
- Ultimate goal: measure the channeling efficiency $\epsilon_{ch} = N_{ch}/N_{tot}$
 - Find N_{tot} , the particles that have entered the crystal with the right angle, so that have the potential to be channeled
 - Find N_{ch}

FINDING N_{tot}

Filtering the data:

1. The particle produce a single track:

preventing from considering events from «pile-up effect»,
electronic noise and delta rays.

2. The beam profile is larger than the crystal. We must select only
particles that physically enter the crystal area:

we need to know the position of the crystals with respect to
the incoming flux of particles.

FINDING N_{tot}

3. Choosing the particles that enter with the right angle $\pm \frac{1}{2} \theta_L$:

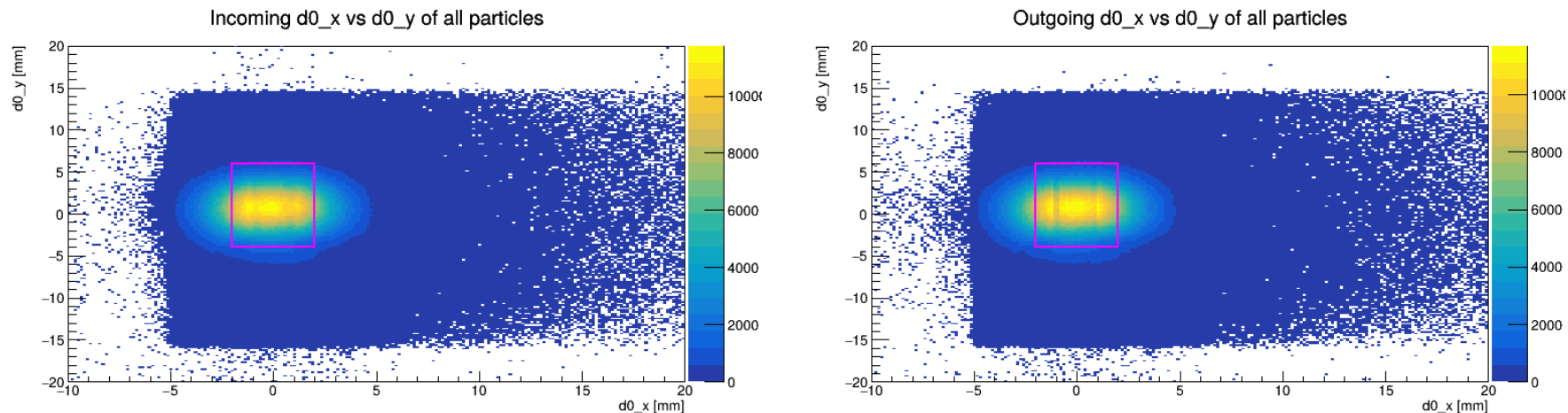
we need to apply a correction θ_0 because the cut may not be centered at zero (the different orientations of the crystallographic planes along the crystal front surface influence the angular range in which particles can be channelled).

4. Choosing a $\theta_0(x, y)$ in order to also apply a torsion correction (because θ_0 actually depends on where the particle hits the crystal in the y direction), which is not negligible in the long crystals.

FILTER 2: SPATIAL CUT

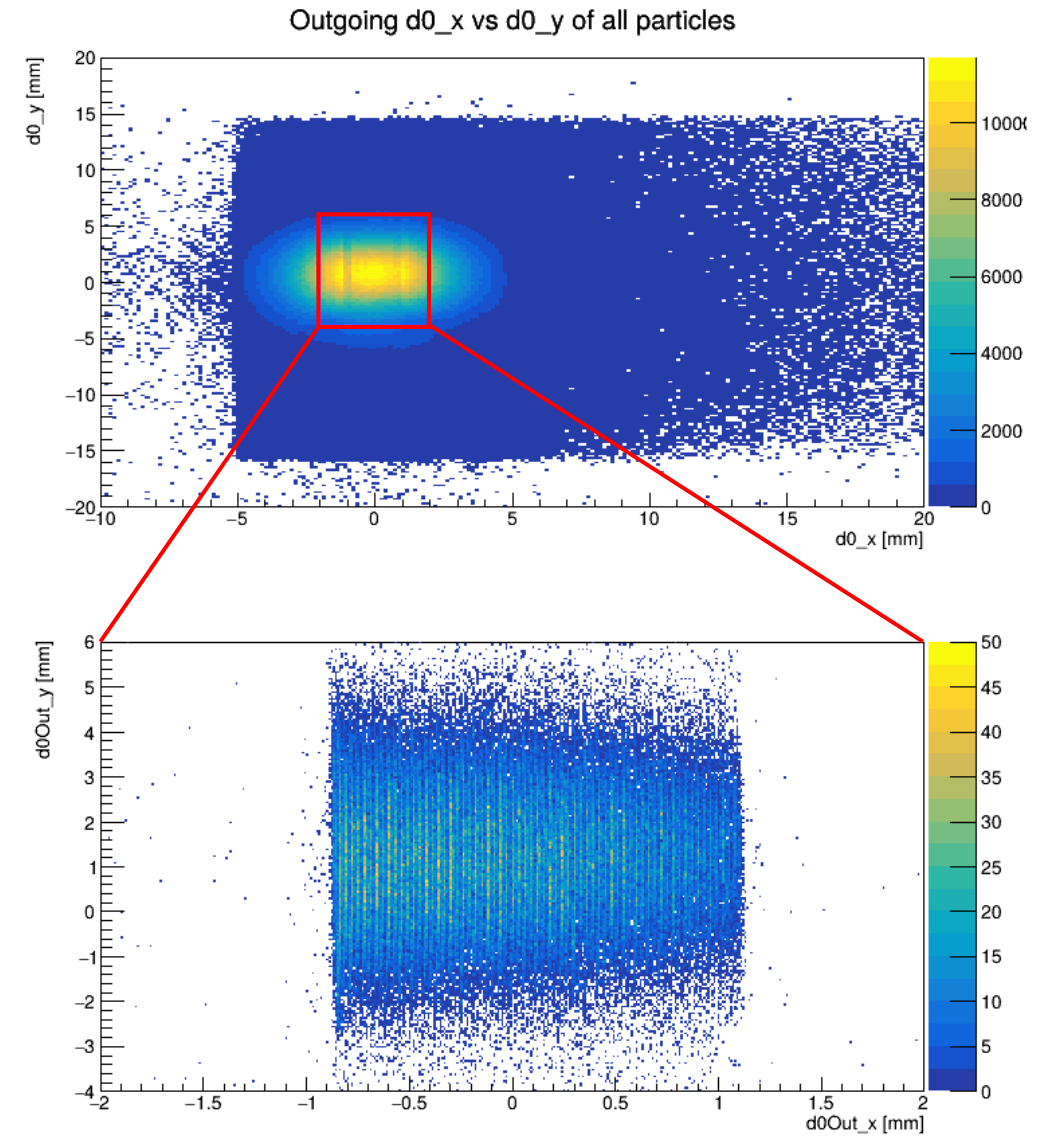
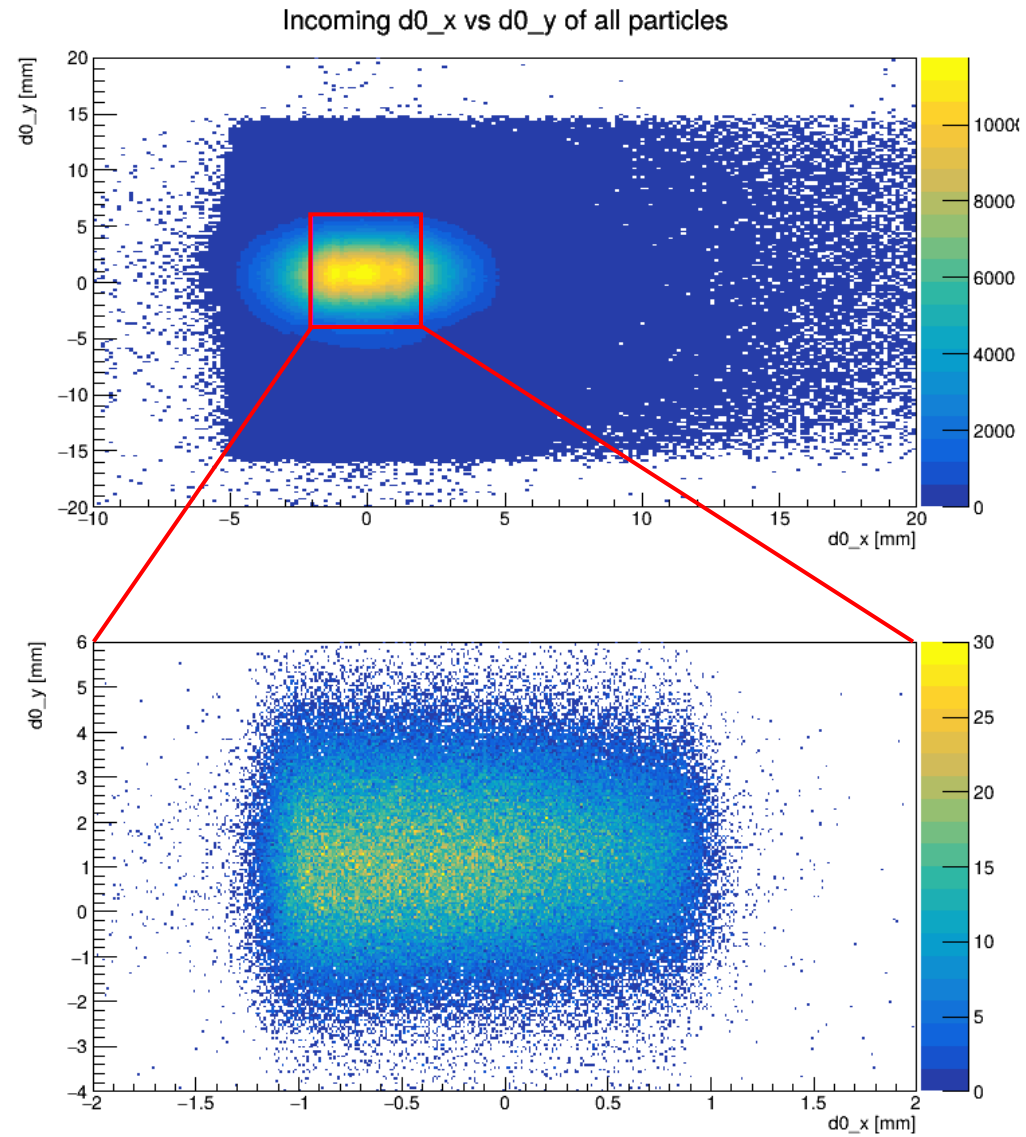
To select the particles that actually enter the crystal we first need to know the position of the crystal with respect to the beam of particles.

Incoming and outgoing streams of particles:

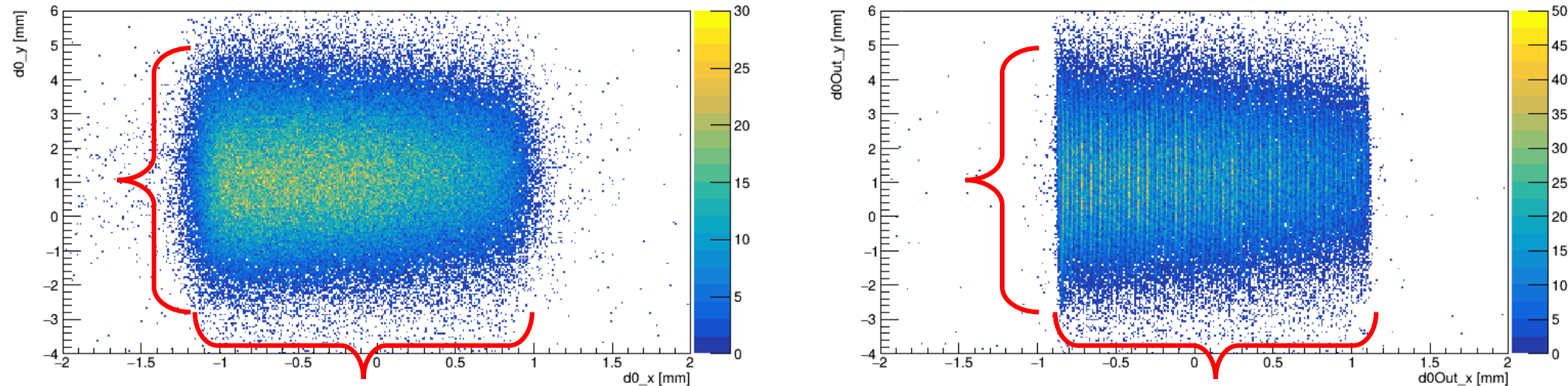


We then look at channeled particles only using a preliminary deflection cut. Their impact points reveal the crystal's physical location.

FILTER 2: SPATIAL CUT



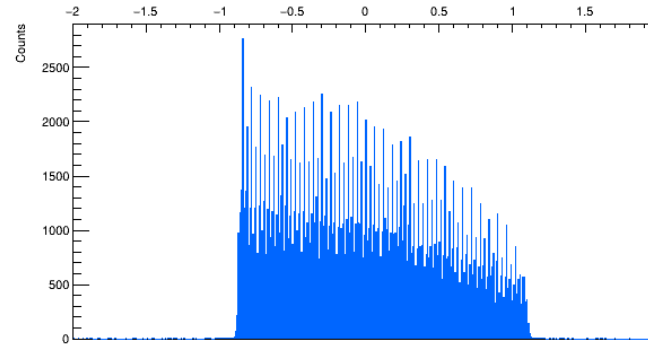
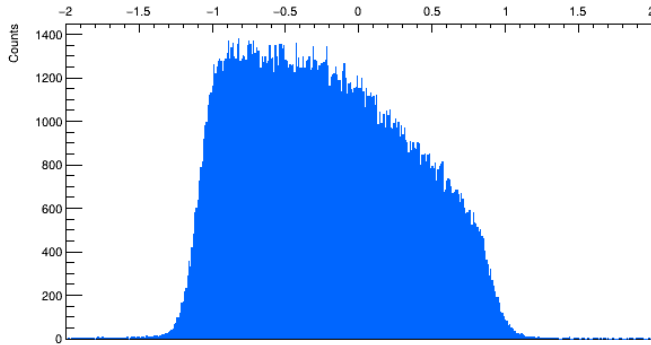
FILTER 2: SPATIAL CUT



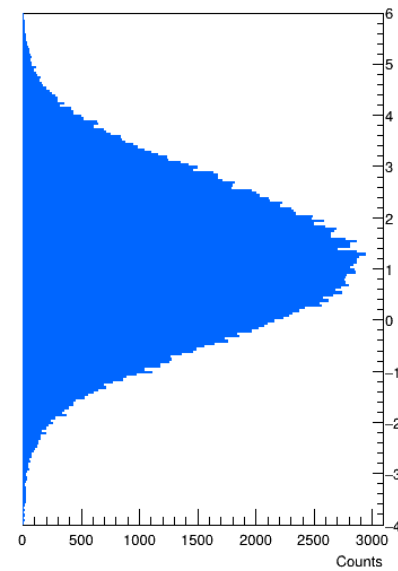
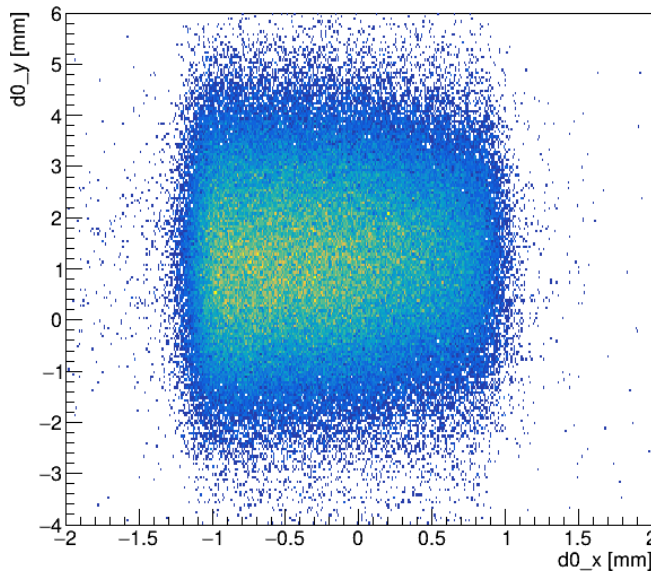
We notice that the position in the beam of the channeled particles has the dimension of the crystal (2mm x 8mm).

To find the edges along x (that have an asymmetrical distribution) we perform a «sliding window» method (fixed width & precision of 0.01mm), aiming for the position that maximize the number of particles inside.

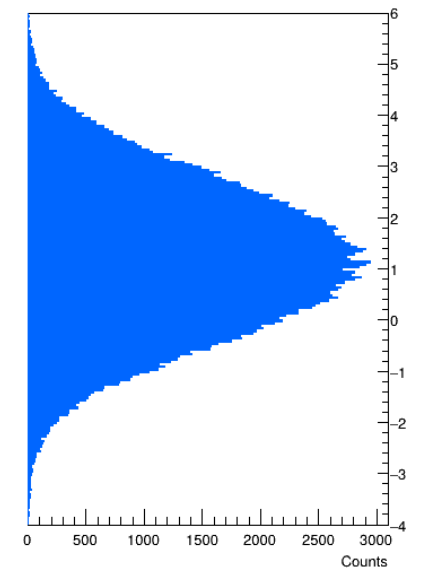
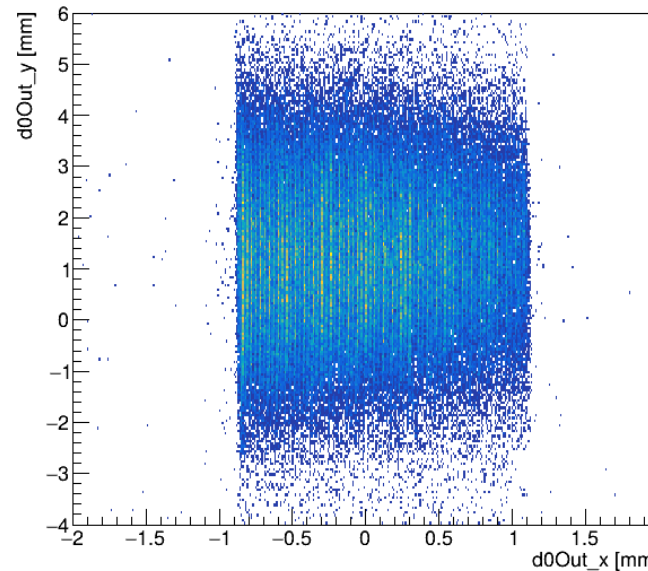
FILTER 2: SPATIAL CUT



The sliding window method was performed on the outgoing distribution, since presents more defined edges.

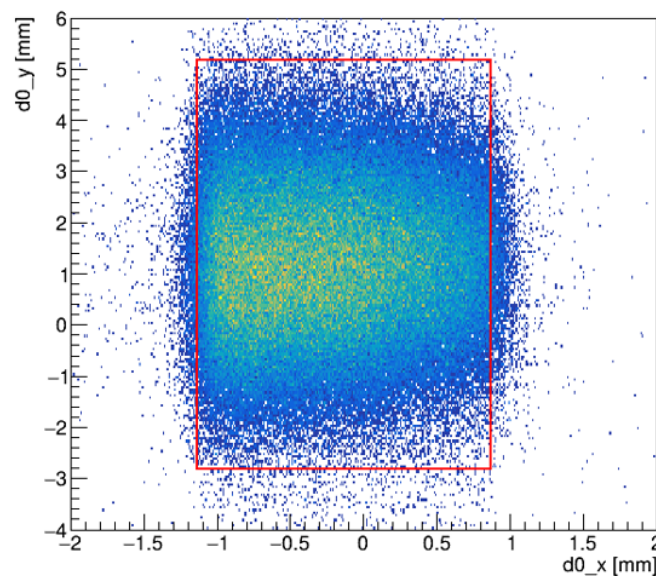
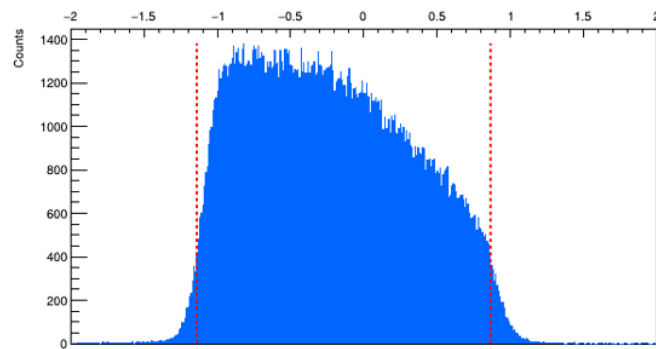


INCOMING BEAM

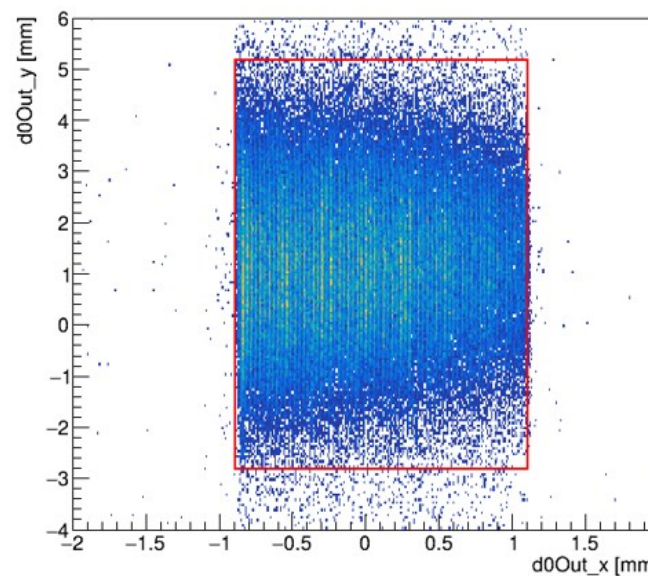
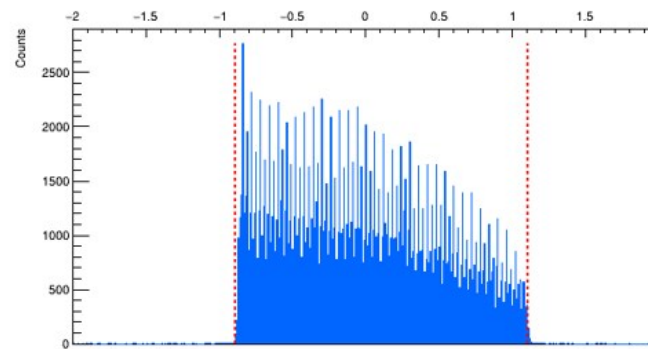


OUTGOING BEAM

FILTER 2: SPATIAL CUT

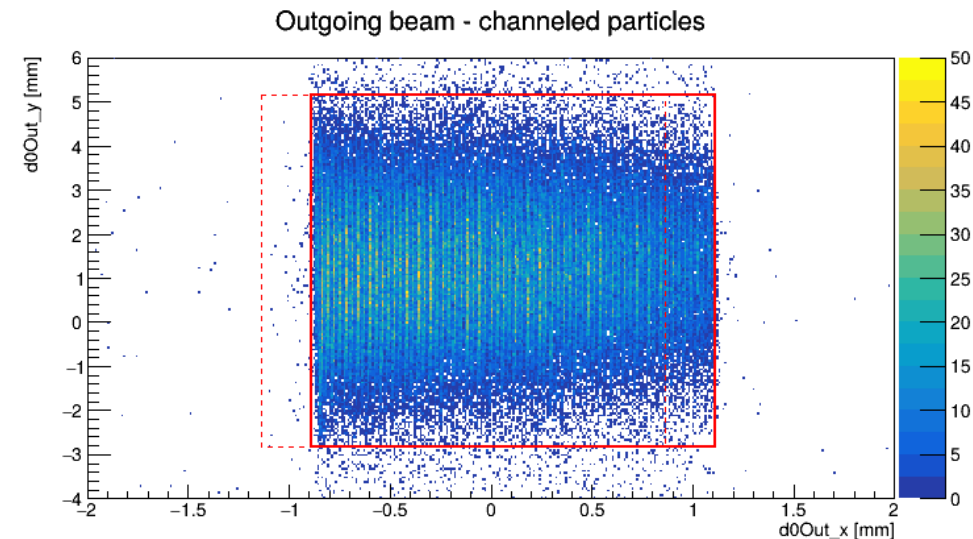
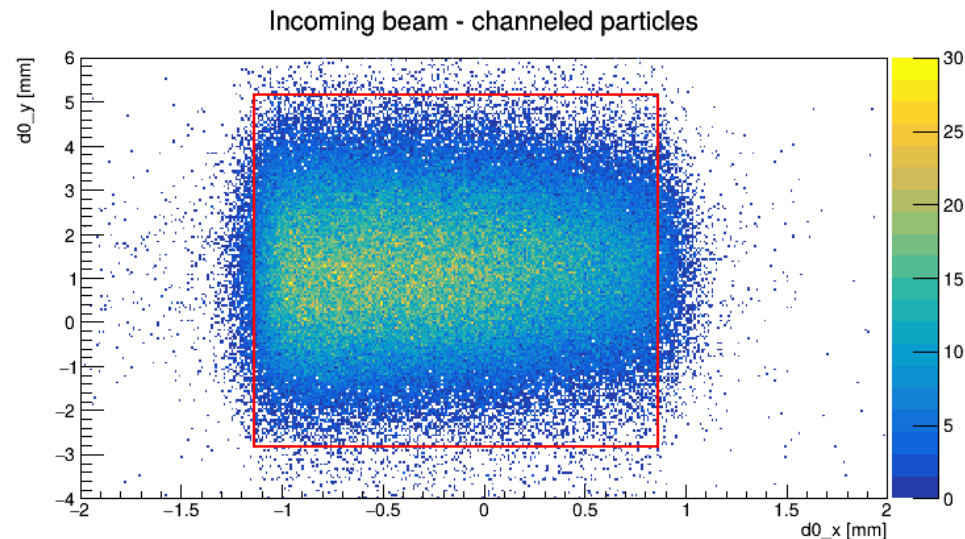


INCOMING BEAM



OUTGOING BEAM

FILTER 2: SPATIAL CUT

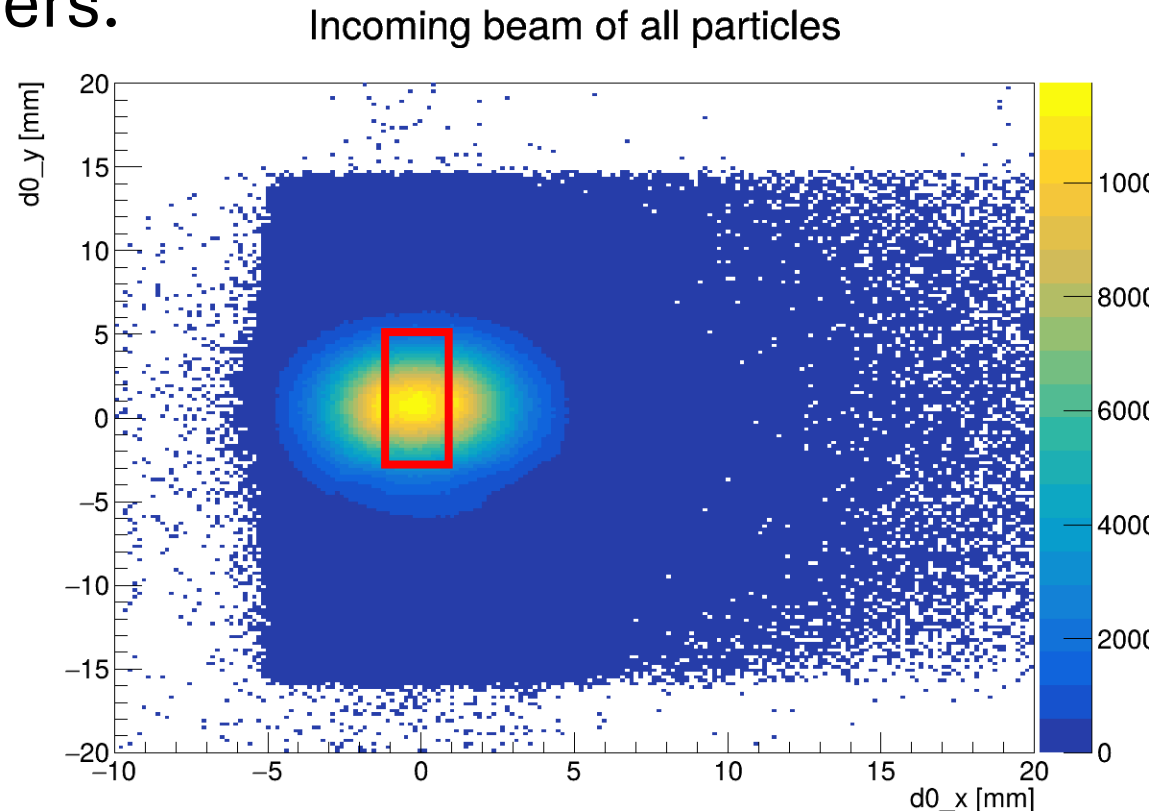


Due to the drift of the channeled particles the window is shifted of ~ 0.24 mm.

This value might not be exact! To make the computation more accurate we need the distance between the detectors (for example).

FILTER 2: SPATIAL CUT

This 2D beam profile ($d0_x$ vs $d0_y$) shows the selected crystal area in red. This spatial filter ensures we only analyze particles with correct impact parameters.



FILTER 3: ANGULAR SELECTION

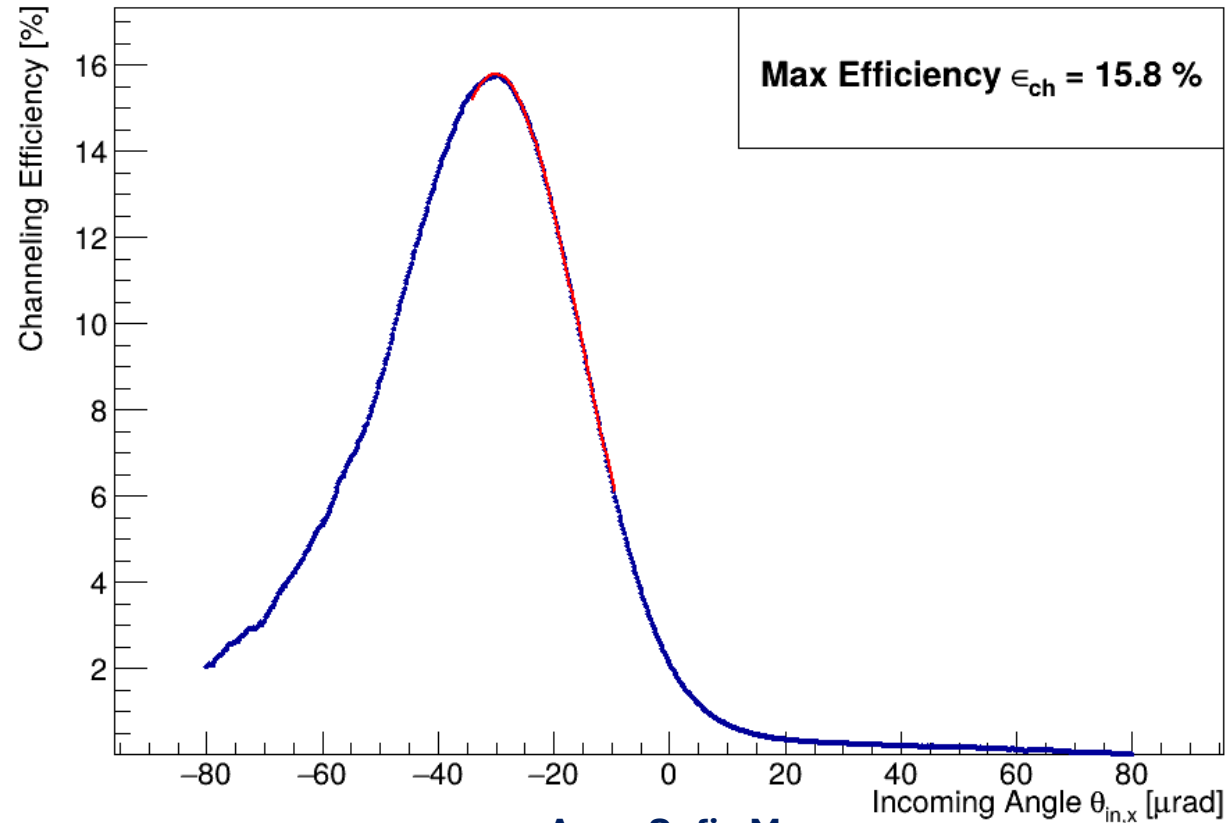
We perform a scan to find θ_0 : the angle where the crystal planes are perfectly aligned with the beam.

- variation of θ_0 along the $\theta_{in,x}$ axis with a precision of μrad
- with every value of θ_0 we select the particles that have entered with the right angle, so $|\theta_{in} - \theta_0| \leq \frac{1}{2} \theta_L$
- with the selected data we compute $\epsilon_{ch} = N_{ch}/N_{tot}$
 - we select θ_0 that maximizes ϵ_{ch}

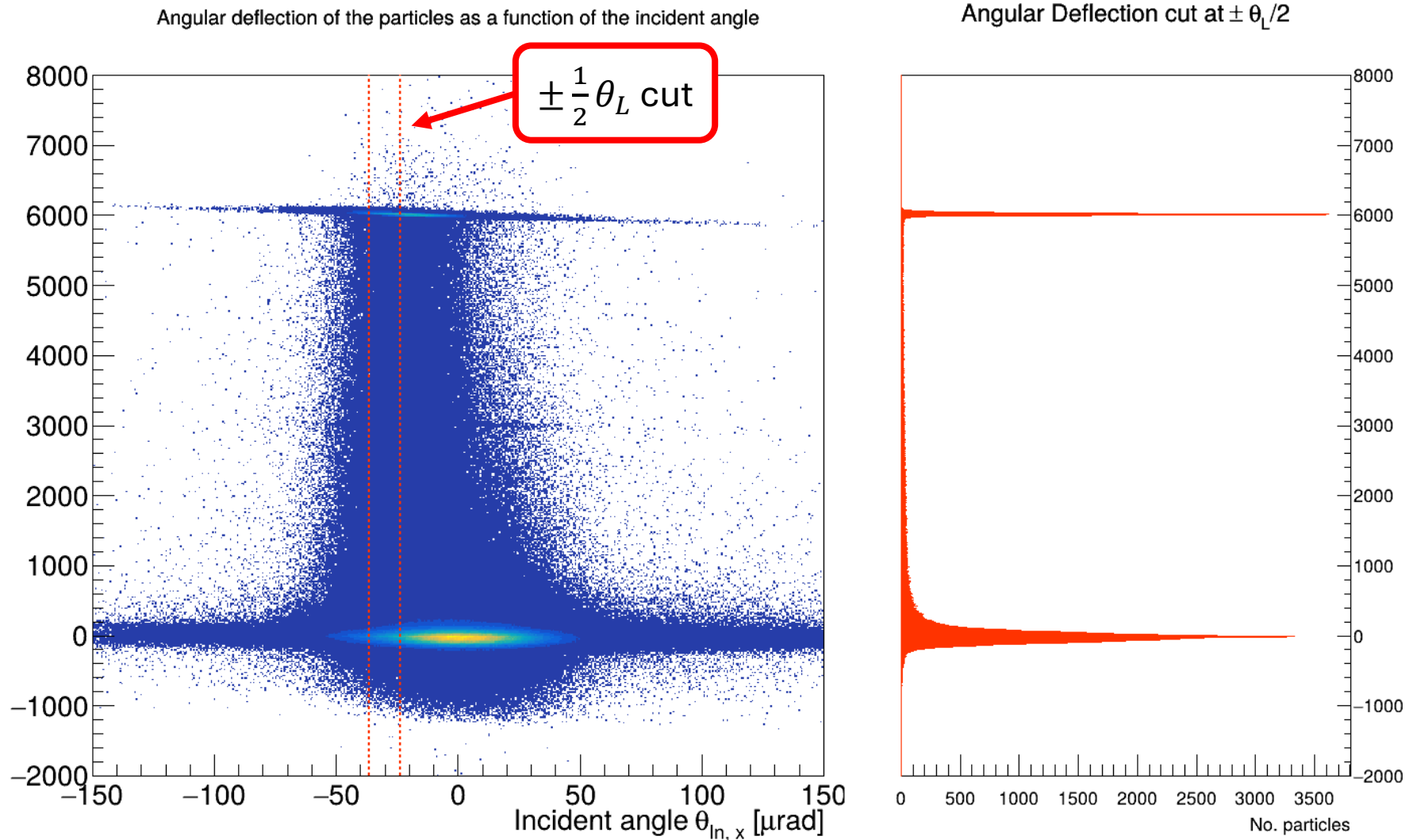
FILTER 3: ANGULAR SELECTION

The efficiency vs incident angle profile is fitted with a Gaussian function. This identifies the mean θ_0 with high precision.

Angular Acceptance Curve

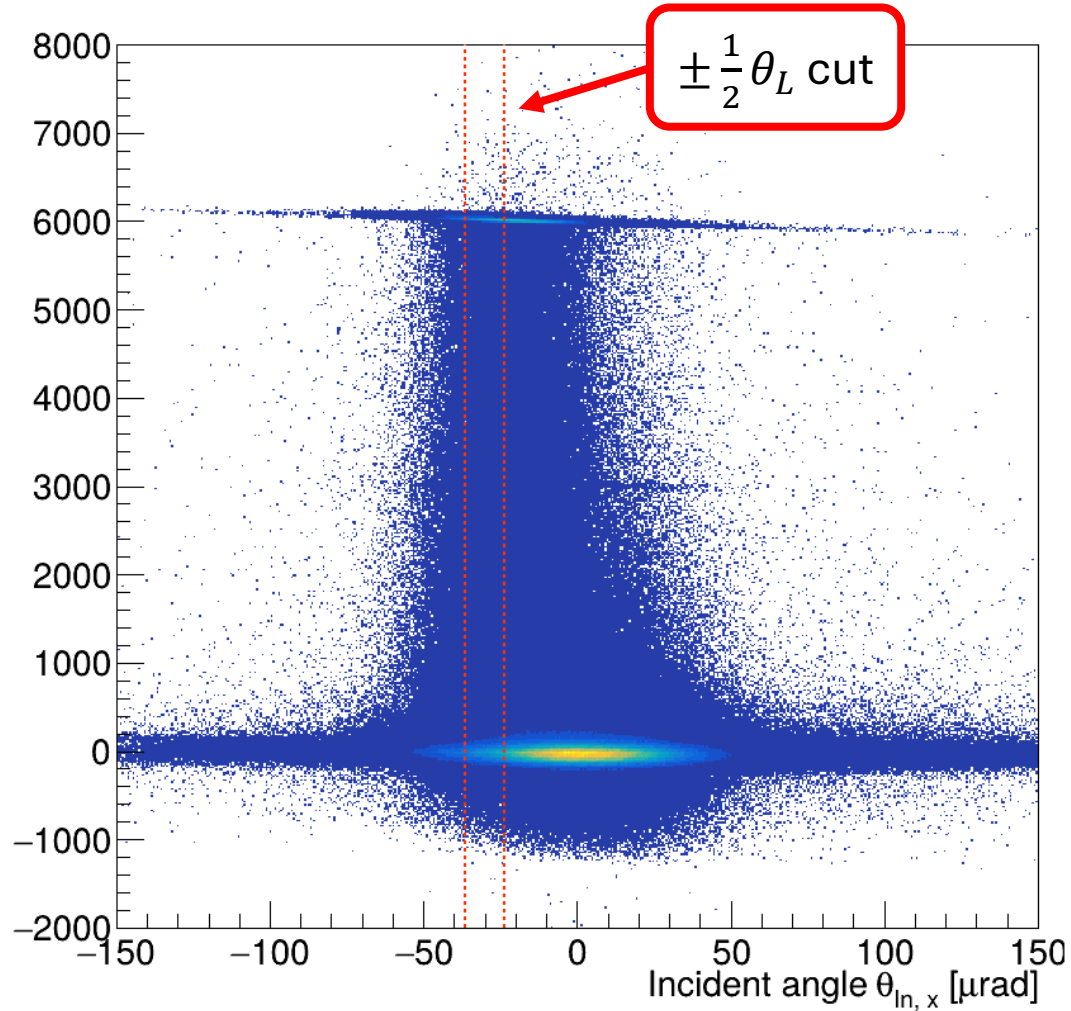


FILTER 3: ANGULAR SELECTION

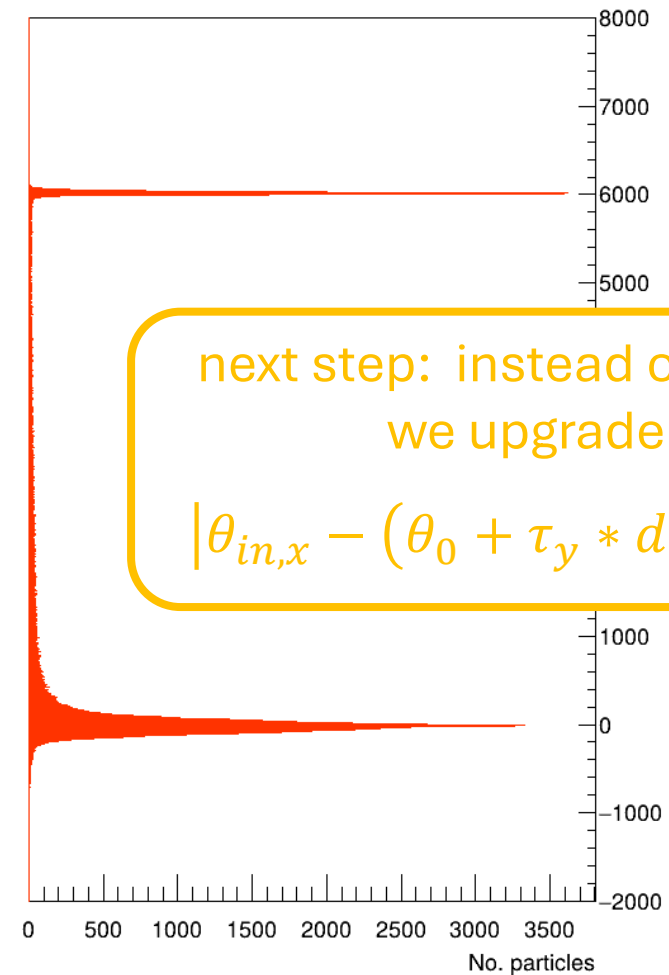


FILTER 3: ANGULAR SELECTION

Angular deflection of the particles as a function of the incident angle

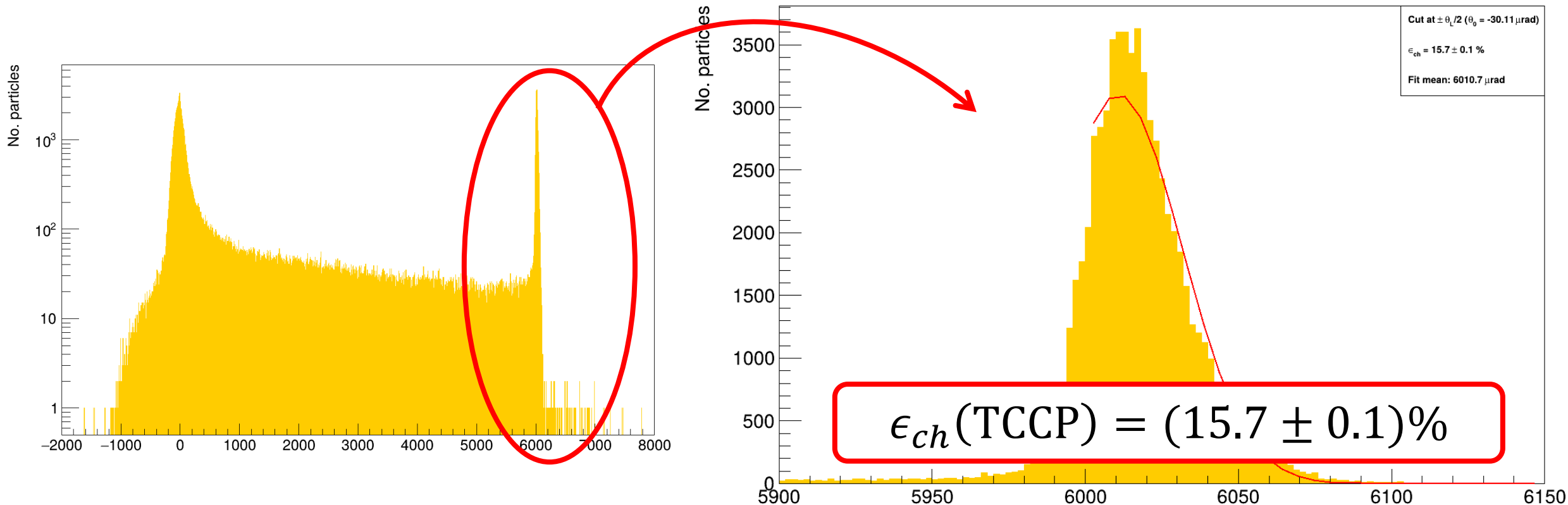


Angular Deflection cut at $\pm \theta_L/2$



MEASUREMENT EXTRACTION

- A Gaussian fit is performed on the deflection peak. We fit the right side of the peak to ensure the cleanest measurement.



TO DO

- ✓ Improve the Δx of spatial shift from incoming and outgoing fluxes of particles by knowing the distance between the two detectors (in the spatial cut)
- ✓ Take into account the torsion effect on the crystals, computing a $\theta_0(x, y)$ by performing a «grid» analysis
 - ✓ Compute the torsion contribution τ_y (variation of the crystalline plane orientation along the direction (y) perpendicular to the bending plane (x))
- ✓ Removing the dechanneling background from the deflection peak (Crystal Ball function?)
- ✓ Perform the analysis for the other 4 datasets (for TCCP and TCCPA) also using them together to have more statistics